

Cyber Security Project 4

System Security Audit & Vulnerability Assessment Report

Submitted by:

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Decode Lab Internship

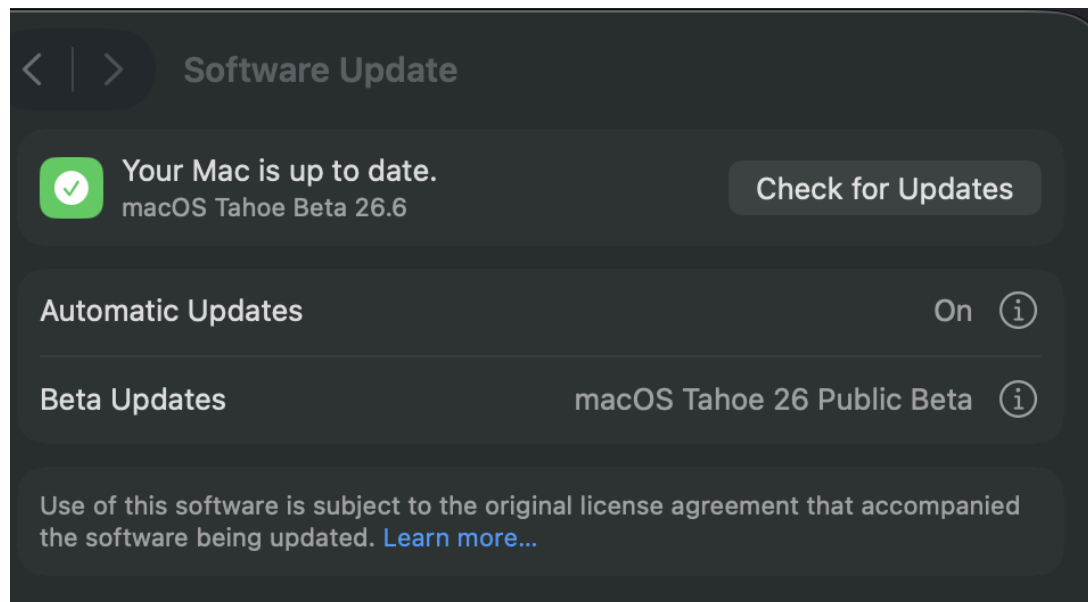
June 2026

Introduction:

The purpose of this project is to assess the security of my personal MacBook by reviewing its current security settings and identifying any potential vulnerabilities. A system security audit was carried out to check important areas such as password protection, software updates, firewall configuration, disk encryption, user account settings, and installed applications.

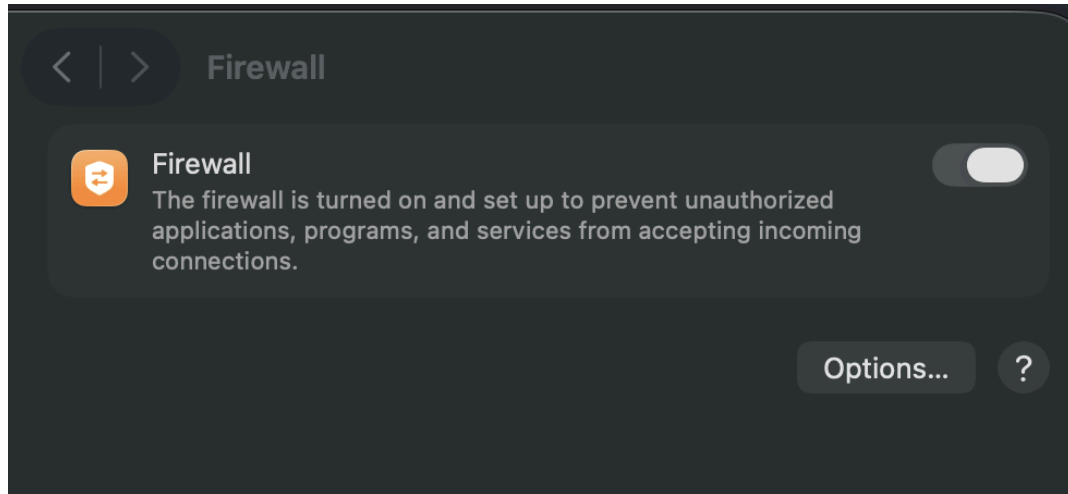
After completing the assessment, any identified issues were reviewed and appropriate security measures were applied where necessary. This project helped me understand the importance of maintaining a secure system and how regular security checks can reduce potential risks and improve overall system protection.

1. Software Update Check:



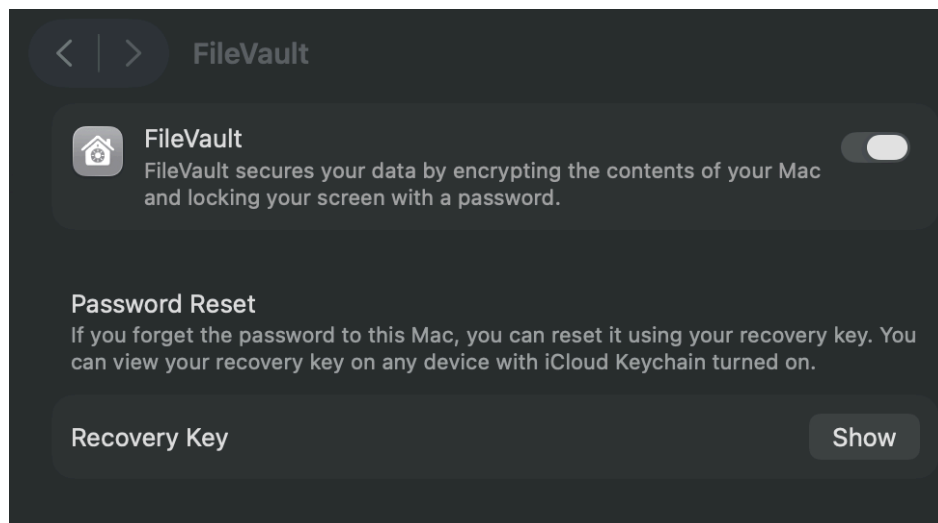
2. Check Firewall

Firewall was reviewed and was found disabled.



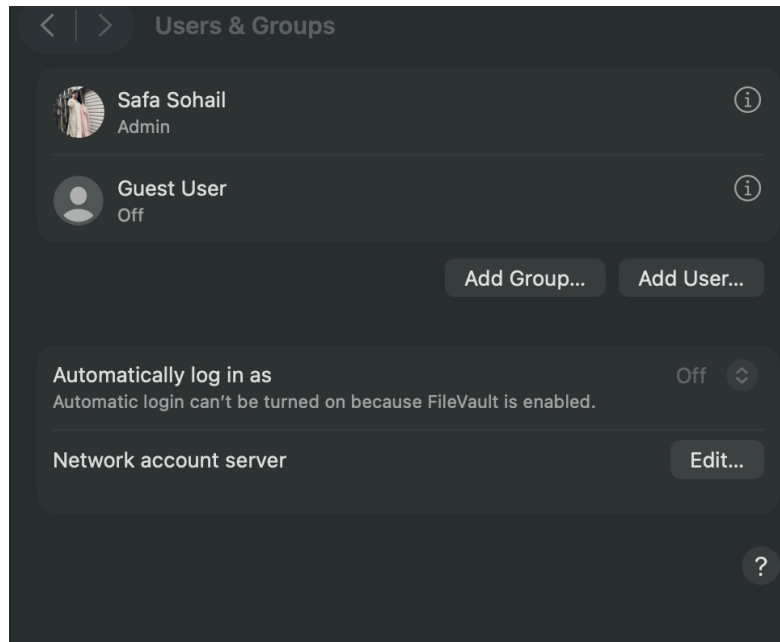
3. Check FileVault

FileVault encryption was enabled to protect the data stored in my macbook

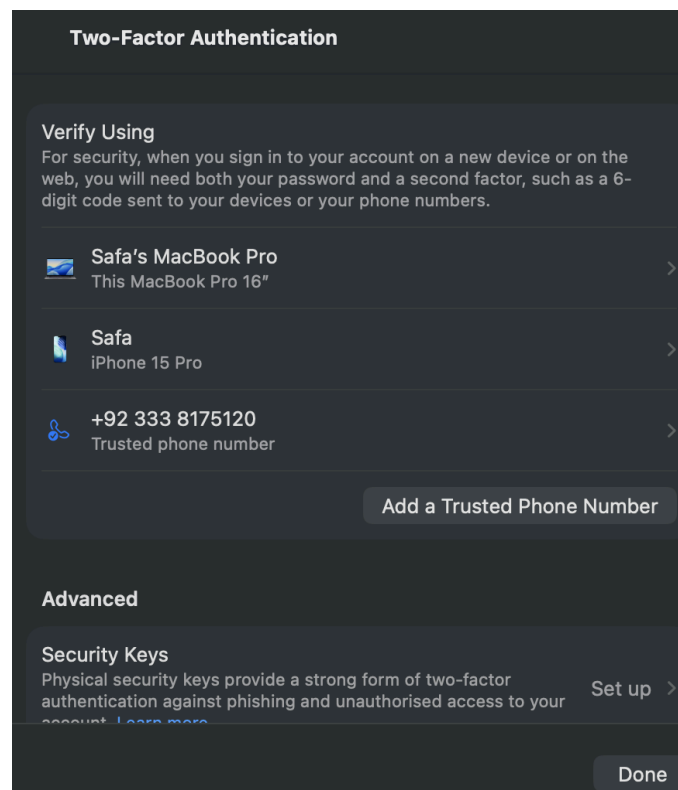


4. Check Guest user

Guest user mode was off



5. Check Two-Factor Authentication



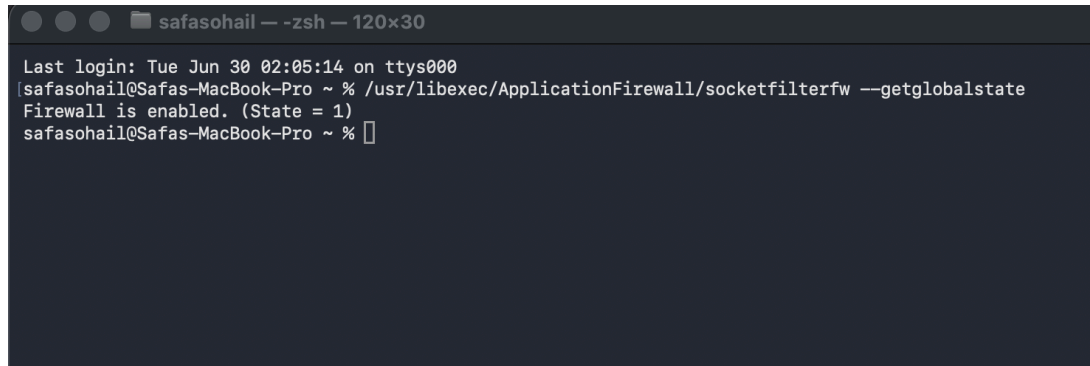
6. Check Installed Apps

All the apps I have installed in my laptop are known and I use most of these apps usually.

7. Check Firewall through terminal

I ran the following command in terminal to see if firewall is enabled or not

`“/usr/libexec/ApplicationFirewall/socketfilterfw --getglobalstate”`

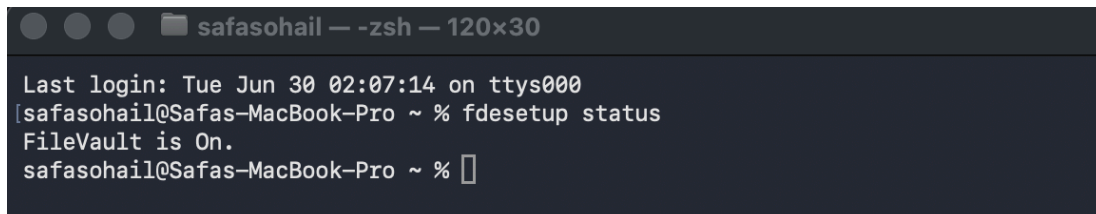
A terminal window titled 'safasohail — -zsh — 120x30' showing the output of the command '/usr/libexec/ApplicationFirewall/socketfilterfw --getglobalstate'. The output indicates that the firewall is enabled (State = 1).

```
safasohail — -zsh — 120x30
Last login: Tue Jun 30 02:05:14 on ttys000
[safasohail@Safas-MacBook-Pro ~ % /usr/libexec/ApplicationFirewall/socketfilterfw --getglobalstate
Firewall is enabled. (State = 1)
safasohail@Safas-MacBook-Pro ~ %
```

8. Check FileVault through terminal

I ran the following command in terminal to see if fileVault is On or Off

`“fdsetup status”`

A terminal window titled 'safasohail — -zsh — 120x30' showing the output of the command 'fdsetup status'. The output indicates that FileVault is On.

```
safasohail — -zsh — 120x30
Last login: Tue Jun 30 02:07:14 on ttys000
[safasohail@Safas-MacBook-Pro ~ % fdsetup status
FileVault is On.
safasohail@Safas-MacBook-Pro ~ %
```

9. Check Software Updates through terminal

```
safasohail — zsh — 120x30

Last login: Tue Jun 30 02:11:03 on ttys000
[safasohail@Safas-MacBook-Pro ~ % softwareupdate -l
Software Update Tool

Finding available software
No new software available.
safasohail@Safas-MacBook-Pro ~ %
```

10. List Installed Apps through terminal

```
Last login: Tue Jun 30 02:21:37 on ttys000
safasohail@Safas-MacBook-Pro ~ % ls /Applications
Apple Configurator.app  Local.app                MAMP PRO.app            Utilities
Google Chrome.app       MAMP                     Safari.app               Visual Studio Code.app
safasohail@Safas-MacBook-Pro ~ %
```

11. Password Strength

A strong password is in use and contains special characters, numbers and alphabets to make it protected from cyber attacks.